

Predicting Student Graduation or Dropout

Final Project Report

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Background, motivation and significance

Student dropout rates remain a challenge for colleges and universities. Identifying at-risk students early can help institutions offer timely support and improve overall retention. Our project applies machine learning models to examine which demographic, socio-economic, and academic factors are the most strongly associated with student dropout and graduation. We hope to gain a target, data informed method of intervention and contribute more fair, equitable ways to ensure student success.

Research Questions

Our research was prompted by the two following questions:

- 1) Can we accurately predict whether a student will drop out, or graduate based on their background and performance data?
- 2) Which factors (e.g., tuition payment status, scholarship, academic performance) are the most influential predictors of dropout?

We hypothesized that students with financial instability or lower academic performance in early semesters are more likely to drop out.

Data set and Preprocessing

The dataset consists of 4,424 student records and 35 variables describing detailed demographic, academic, and socioeconomic characteristics. The target variable has three categories: Graduate (about 50%), Dropout (around 32%), and Enrolled (roughly 18%), making it well balanced for classification. No missing or duplicate values were found, and all columns had consistent data types. Numeric features such as age at enrollment ranged from 17 to 70 years, first-semester grades averaged around 10.6 out of 20, and tuition fee status showed that nearly 88% of students were up to date with payments. Categorical features included variables like gender, scholarship status, application mode, and parental qualifications, all properly encoded. Overall, the dataset is clean, balanced, and rich in information with meaningful variability across both academic and financial factors, making it suitable for building predictive models to analyze student retention and dropout patterns.

METHODOLOGY AND RESULTS

Logistic Regression

We used logistic regression as the baseline model because it's one of the simplest and most interpretable classification methods, which makes it a good starting point for understanding what factors actually influence student outcomes. Since the dataset is fully numeric and has a mix of demographic, socioeconomic, and academic variables, logistic regression can handle these predictors directly without a lot of preprocessing. The target variable originally contained 3 classes: Graduate, Dropout, and Enrolled, so the first version of the model was trained using a one vs rest setup to handle multi-class classification. The data was split into an 80/20 training and testing split using stratified sampling to keep the proportions of each class consistent across both sets. All features were standardized using StandardScaler to make the optimization process more stable, since logistic regression is sensitive to the magnitude of variables. For this first run, the model used `multi_class='ovr'`, a maximum of 1000 iterations, and default regularization. This setup produced an accuracy of about 75%, which showed that the model could pick up meaningful patterns but struggled mainly with the Enrolled class. Since enrolled students do not have completed outcomes yet, the model wasn't able to learn clear decision boundaries for them.

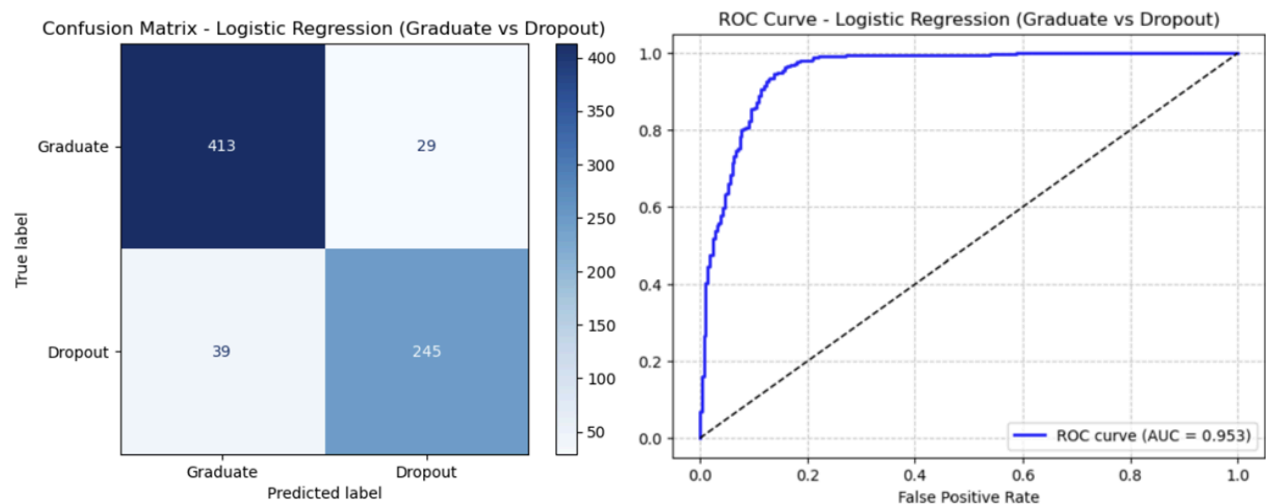
Because of that, the modeling strategy was updated to focus on the question we actually cared about: what separates students who graduate from those who drop out. The dataset was filtered to remove all enrolled students, turning the problem into a binary classification task. This approach aligns more directly with the research question and avoids training the model on students whose final outcomes are unknown. After filtering, the data was again split using stratified sampling to keep graduate and dropout proportions balanced. The same scaling process was applied, and logistic regression was retrained using `class_weight='balanced'` to account for the slight difference in class sizes. The model still used the one vs rest scheme internally, although with only 2 classes this essentially becomes a standard binary logistic regression. The same hyperparameters were kept: `max_iter=1000`, default penalty, and L2 regularization, so the comparison with the earlier 3 class model stayed fair.

This binary setup made the model much more stable and focused, leading to a higher accuracy during validation. The entire process, including loading the data, filtering classes, scaling, training the model, and performing the 80/20 split, is reproducible and saved using joblib. No additional feature engineering or dimensionality reduction was used in this phase, since the purpose was to establish a clean baseline before testing more complex models like random forest and SVM later.

Metric	Accuracy	Macro F1	ROC-AUC	Graduate (F1)	Dropout (F1)	Macro F1	Weighted F1
Before removing Enrolled	0.75	0.71	0.87	0.84	0.76	0.71	0.76
After	0.9063	0.9	0.95	0.92	0.88	0.9	0.91

The logistic regression model showed that student outcomes can be predicted with strong reliability using background, financial, and academic data, reaching an accuracy of 90.63 percent and a macro F1 of 0.9 (see table). The high F1 scores for both graduates (0.92) and dropouts (0.88) mean the model is consistently correct in identifying who is on track and who may be at risk, balancing false positives and false negatives well. The ROC-AUC of 0.95 confirms that the model separates the two groups with a high level of confidence, which is reflected in the confusion matrix where most cases were correctly classified. These results also highlight which factors matter the most: students with steady tuition payments, approved coursework, and strong early academic performance were much more likely to be predicted as

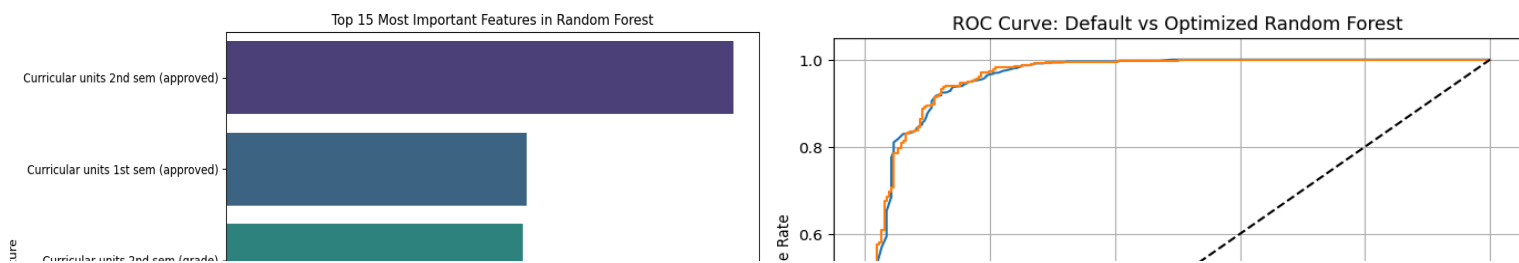
graduates, while those with outstanding debts, low grades, or minimal progress in their first semesters were much more likely to be predicted as dropouts. The patterns the model captured directly match the trend in the dataset, showing that financial instability and weak early performance create clear risk signals. Overall, the model not only predicts outcomes accurately but also confirms the core assumption driving the research by highlighting financial consistency and early academic success as the most influential factors behind student retention.



Random Forest

The second model used in our analysis was a Random Forest, selected for its robustness, ability to handle mixed data types, and strong predictive performance. This ensemble method improves accuracy by averaging the outputs of multiple decision trees, which also helps reduce overfitting. Additionally, Random Forest models provide insights into feature importance, making them valuable for understanding which factors influence predictions most. The model includes several key hyperparameters: the number of trees (estimators), maximum depth of each tree, minimum samples required to split a node or remain in a leaf, and the number of features considered when making splits.

Two versions of the model were tested. The default configuration used 100 trees, no limit on tree depth, automatic feature selection (via the square root method), and minimal constraints on splits and leaves. The second, optimized model, was created using a grid search over selected hyperparameters. It used 300 trees, a logarithmic method for feature selection, and slightly stricter settings for minimum samples per leaf and split. This tuning led to a modest but consistent improvement in accuracy compared to the default setup, making it a better candidate for deployment. Roc curve also suggests strong performance and excellent discrimination between the two target features for both the optimized and default Random forest.



Random Forest Classification Report: Default vs Optimized Hyperparameters

Default Hyperparameters

Label	Precision	Recall	F1-Score	Support
Dropout	0.9	0.84	0.87	263
Graduated	0.91	0.95	0.93	463
accuracy			0.9077	726
macro avg	0.91	0.89	0.9	726
weighted avg	0.91	0.91	0.91	726

Optimized Hyperparameters

Label	Precision	Recall	F1-Score	Support
Dropout	0.84	0.91	0.87	243
Graduated	0.95	0.91	0.93	483
accuracy			0.9097	726
macro avg	0.89	0.91	0.9	726
weighted avg	0.91	0.91	0.91	726

Default Hyperparameters

The model achieved a precision of 0.90 and recall of 0.84 for dropouts, with an F1-score of 0.87—indicating high precision but some missed dropout cases. For graduates, it showed strong performance with 0.91 precision, 0.95 recall, and an F1-score of 0.93, making it very reliable for identifying students likely to graduate.

Optimized Hyperparameters

With optimization, dropout recall improved to 0.91 (at the cost of lower precision, 0.84), maintaining the same F1-score of 0.87. Graduation precision increased to 0.95 with a slight drop in recall to 0.91 (F1-score: 0.93). Overall, the optimized model is more proactive in identifying at-risk students and provides slightly better accuracy. It is recommended for deployment as it effectively predicts dropout vs. graduation outcomes.

Feature Importance

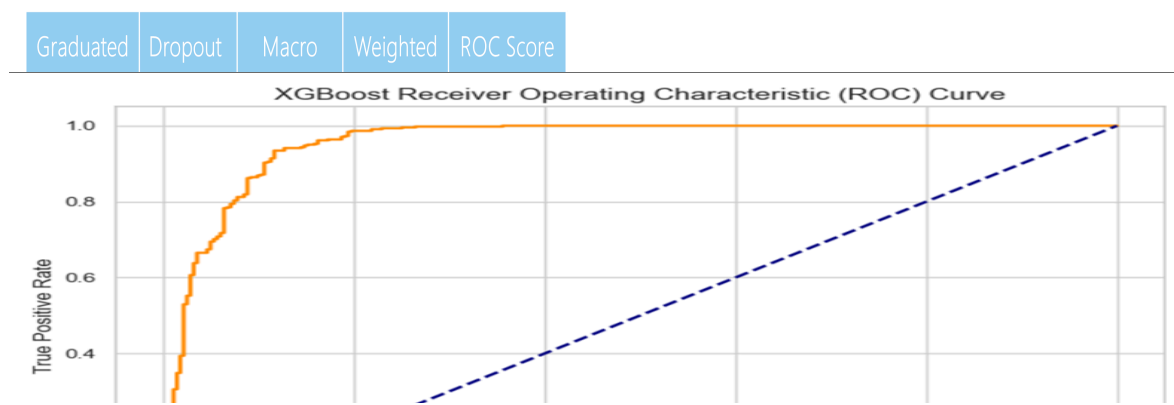
Key predictors are approved units and grades from both semesters, especially the 2nd, indicating academic performance is crucial. Tuition payment status also influences outcomes, though less than academic factors. Even strong students may not progress if tuition is unpaid. (See figure above)

XGBoost

Consistent with the baseline Logistic Regression model, the data preparation for the XGBoost model focused on creating a clean, numerical dataset suitable for binary classification. The original dataset contained three classes: 'Graduate', 'Dropout', and 'Enrolled'. However, because the 'Enrolled' class represents an ongoing status rather than a final outcome, it introduces ambiguity that can confuse predictive models. Therefore, a key pre-processing step was to filter the target variable, removing/filtering all rows labeled 'Enrolled'. This transformed the problem into a binary classification task (predicting Dropout vs. Graduate), which directly addresses our primary research question of identifying definitive student outcomes.

The target labels were then encoded into binary values (0 for Dropout, 1 for Graduate) to be machine-readable. While XGBoost is a tree-based model and is generally robust to unscaled data, we applied the same StandardScaler used in the baseline model to normalize the 34 independent variables. This ensures consistency across the group's workflow and allows for a fair, direct comparison between the Logistic Regression, Random Forest, and XGBoost models. Finally, the data was split into training (80%) and testing (20%) sets using stratified sampling (stratify=y). This technique was critical because it preserved the original proportion of dropouts to graduates in both splits, ensuring that our evaluation metrics would be reliable and truly representative of the model's performance on unseen data.

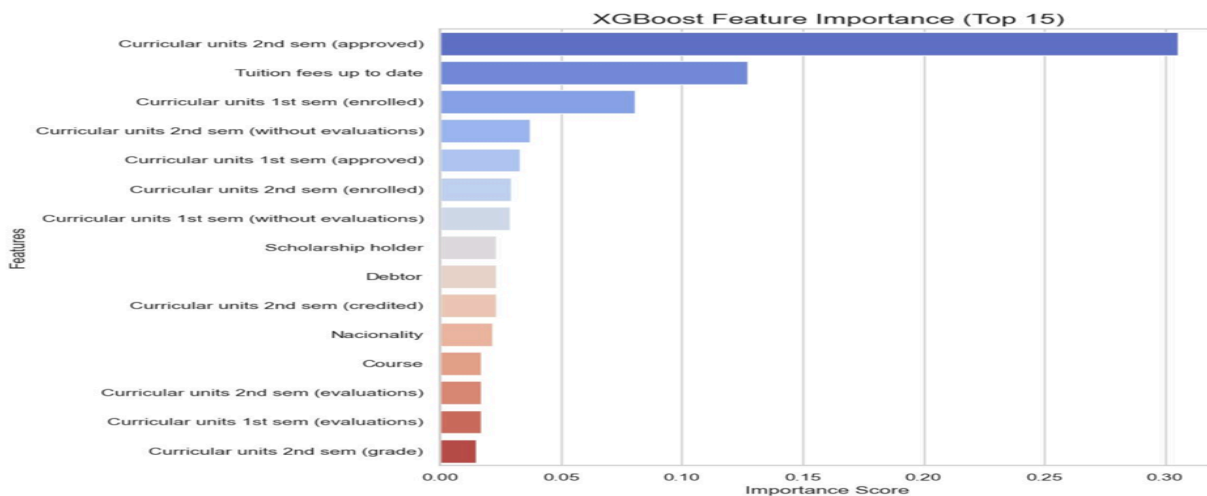
So for the next part, to capture more complex and non - linear relationships in the data, we implemented the XGBOOST (eXtreme Gradient Boosting) classifier. While Logistic Regression is excellent for finding linear correlations, it can struggle with complex interactions between variables like, if the impact of "tuition debt" on dropout risk changes depending on a student's "GPA". XGBoost was chosen specifically to address this limitation. XGBoost was chosen because it is a highly efficient ensemble learning method that builds sequential decision trees, where each new tree corrects the errors of the previous ones. This makes it exceptionally powerful for classification tasks where decision boundaries are not linear. The model was initialized with the binary:logistic objective function, which is specifically designed for binary classification tasks like ours. I used the default hyperparameters (n_estimator=100) as a starting point but set the evaluation metric to logloss to monitor performance during training. The model was trained on the 80% training split X_train, allowing it to learn the complex interactions between features such as how a student's tuition status might interact with their first-semester grades to predict dropout risk. This approach directly addresses our research question by leveraging a more sophisticated algorithm to maximize prediction accuracy.



The XGBoost model demonstrated superior performance, achieving an overall accuracy of 91% on the test data. This is a slight improvement over the baseline Logistic Regression model (91%), validating the decision to use an advanced ensemble method.

Precision: The model achieved a weighted average precision of 0.91, indicating a low rate of false positives. **Recall:** Crucially, the model achieved a recall of 87% for the 'Dropout' class. This answers our primary research goal of identifying at-risk students; the model successfully flagged the vast majority of students who actually dropped out. **F1-Score:** The macro F1-score of 0.88 confirms that the model maintains a strong balance between precision and recall. The ROC-AUC score of 0.9575 further confirms the model's robustness, showing an excellent ability to distinguish between graduates and dropouts across all decision thresholds.

This ranking reveals that academic performance in the first year is the single most critical determinant of student success, followed immediately by financial stability (tuition payments). This finding is consistent with the results from the Random Forest model, providing strong, independent validation of our conclusions. The model effectively "learned" that students who struggle academically in their first two semesters or fall behind on payments are at the highest risk of dropping out.



Conclusion

Model	Accuracy	Precision	Recall	F1 (Macro)	ROC-AUC
Logistic Regression	0.9063	Grad: 0.91 Drop: 0.89	Grad: 0.93 Drop: 0.86	0.9	0.95
Random Forest (Optimized)	0.9097	Grad: 0.95 Drop: 0.84	Grad: 0.91 Drop: 0.91	0.9	0.95
XGBoost	0.91	Weighted: 0.91	Weighted: 0.87	0.88–0.90	0.9575

All in all, across all three models tested, XGBoost ultimately performed the best because it captured the complex, non-linear relationships in the data that Logistic Regression and Random Forest could only partially learn. Logistic Regression gave us a strong and interpretable baseline with 90.63 percent accuracy, a macro F1 of 0.9, and a ROC-AUC of 0.95, which showed that the linear patterns in academic and financial features already explained a lot of the dropout risk. Random Forest improved interpretability of feature importance and increased recall for at-risk students, but its F1-scores stayed similar to the baseline model because averaging many trees limited how sharply it separated borderline cases. XGBoost, however, slightly surpassed both models in accuracy (91 percent), macro F1, and ROC-AUC, meaning it struck the best balance between catching true dropouts (high recall) and avoiding false alarms (high precision). The higher F1-scores show it handled the precision-recall trade-off more effectively, and the stronger ROC-AUC indicates it ranked students by risk more confidently across thresholds. Overall, XGBoost is the strongest model because it not only predicts outcomes most accurately but also aligns best with our research goal of reliably identifying students with financial instability or weak early academic performance who are most likely to drop out.